Appendix A - Eastern Recherche Marine Park Natural Values Report

South-west Marine Parks Network

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Natural Values

Understanding of natural values

Benthic ecosystem extent

Observations

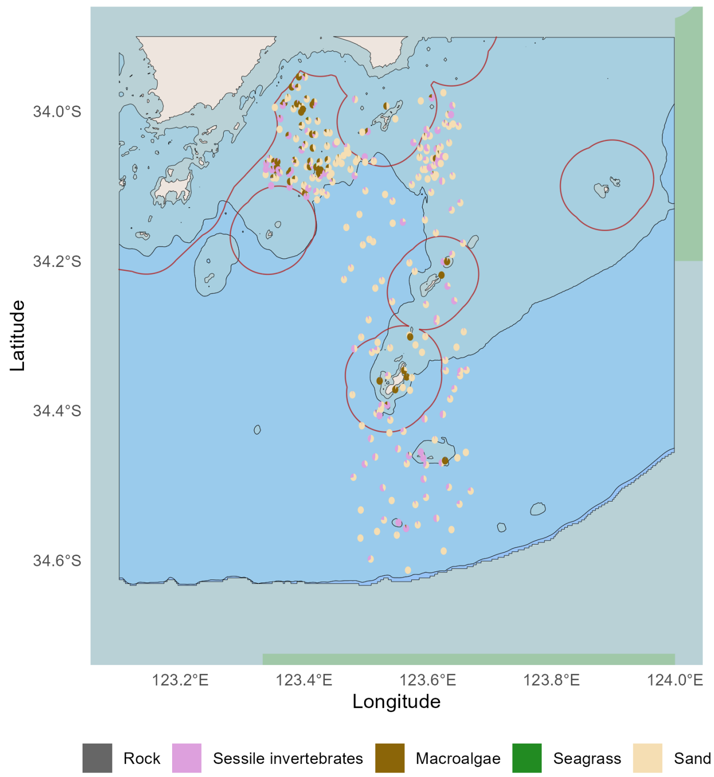

Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera (BOSS) imagery visualised as spatial pie charts, pooled across the 2022 and 2025 surveys. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

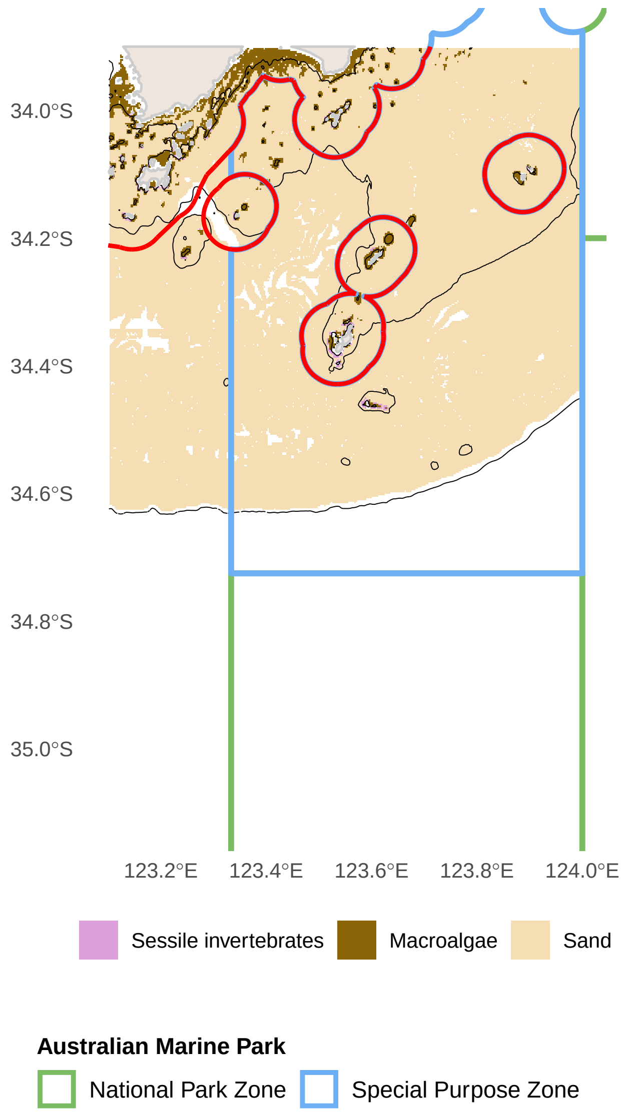

Figure 2: Predicted dominant habitat type from stereo-BRUV and drop-camera groundtruthing, pooled across survey years. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

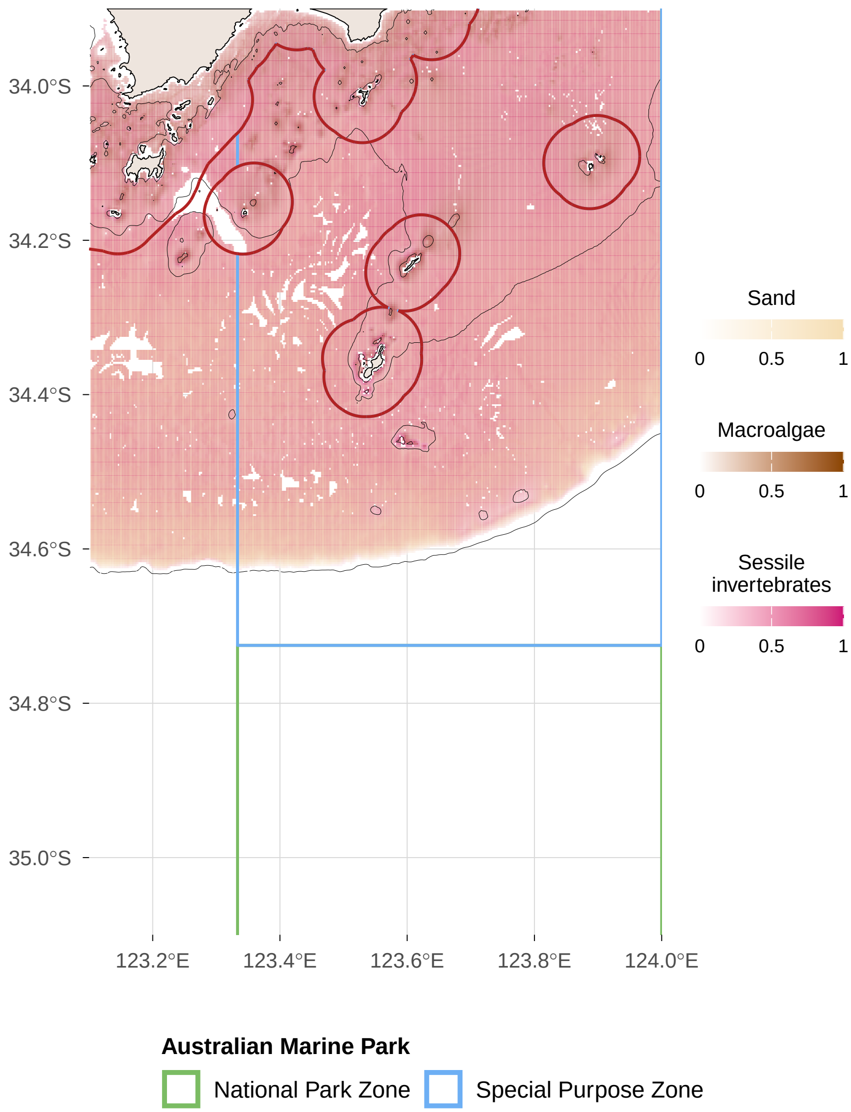

Figure 3: Predicted dominant benthic habitat from stereo-BRUV and drop-camera groundtruthing, pooled across survey years. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

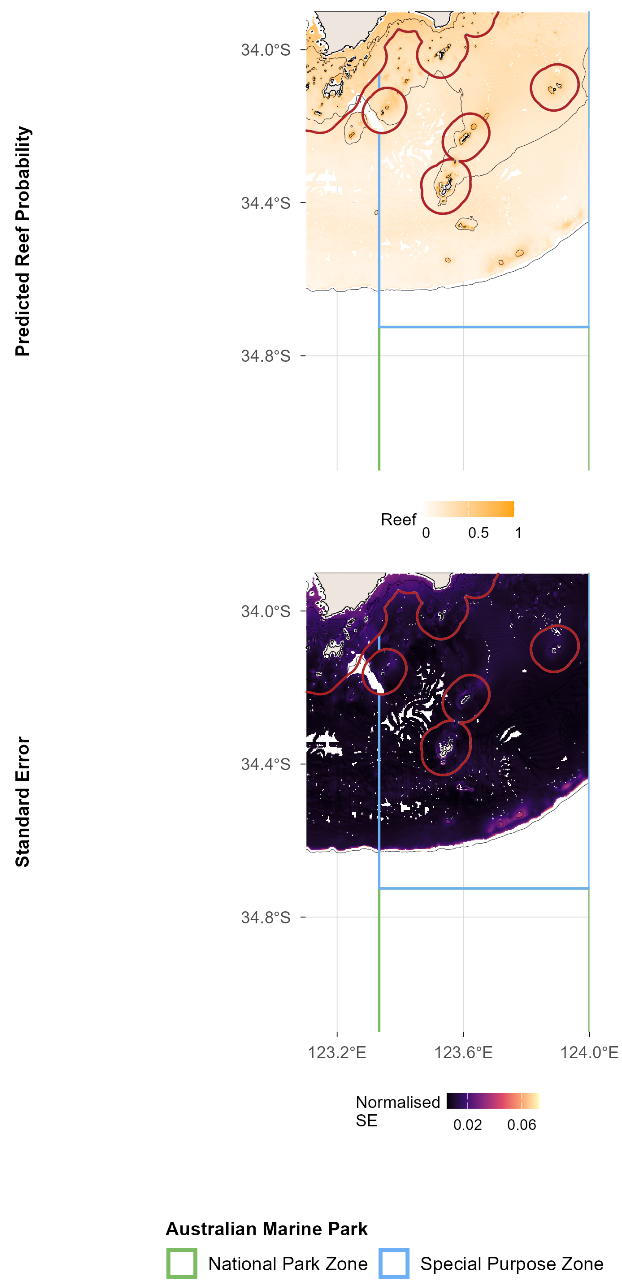

Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across survey years. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

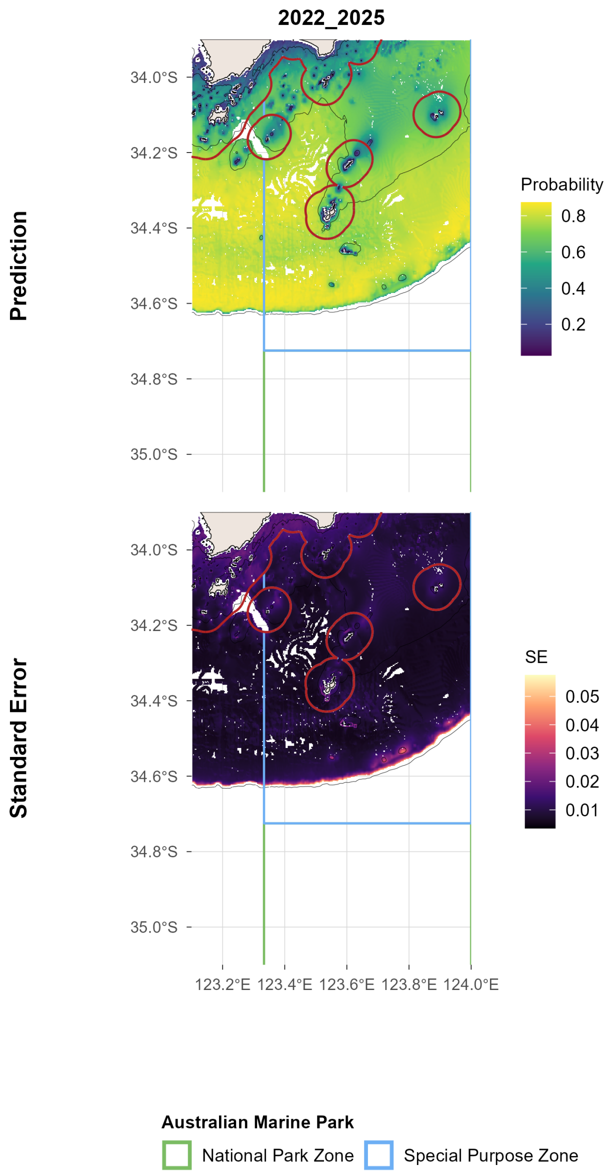

Figure 5.1: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across survey years. ERMP = Eastern Recherche Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

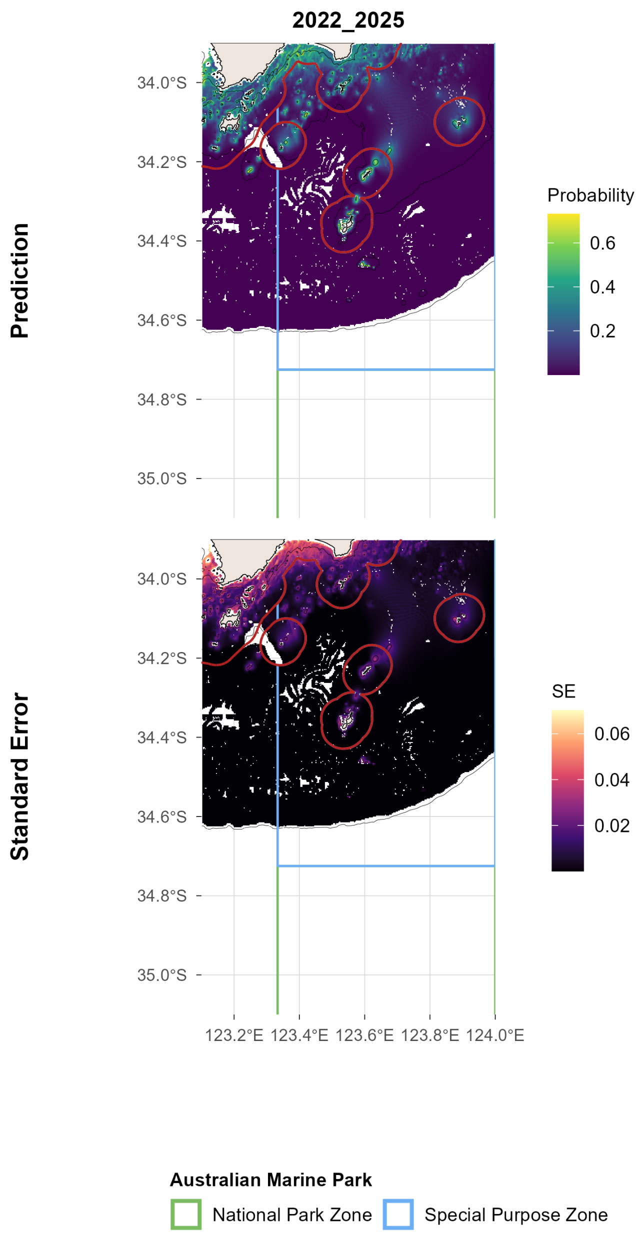

Figure 6.1: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across survey years. ERMP = Eastern Recherche Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

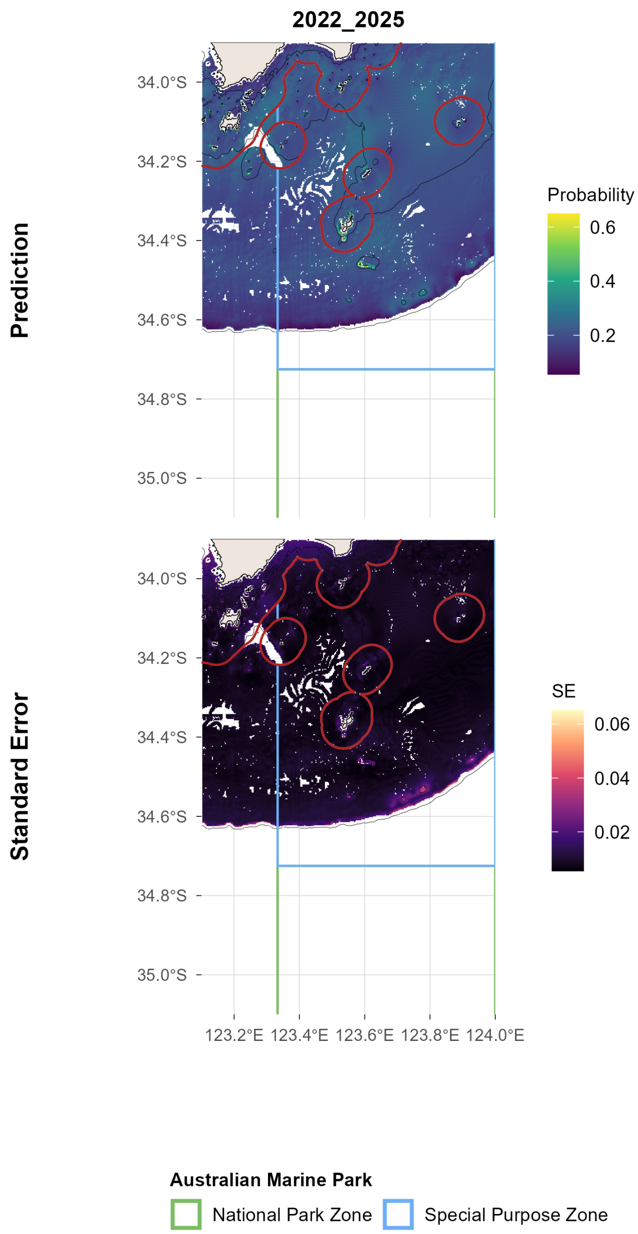

Figure 7.1: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV and drop-camera groundtruthing, pooled across survey years. ERMP = Eastern Recherche Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

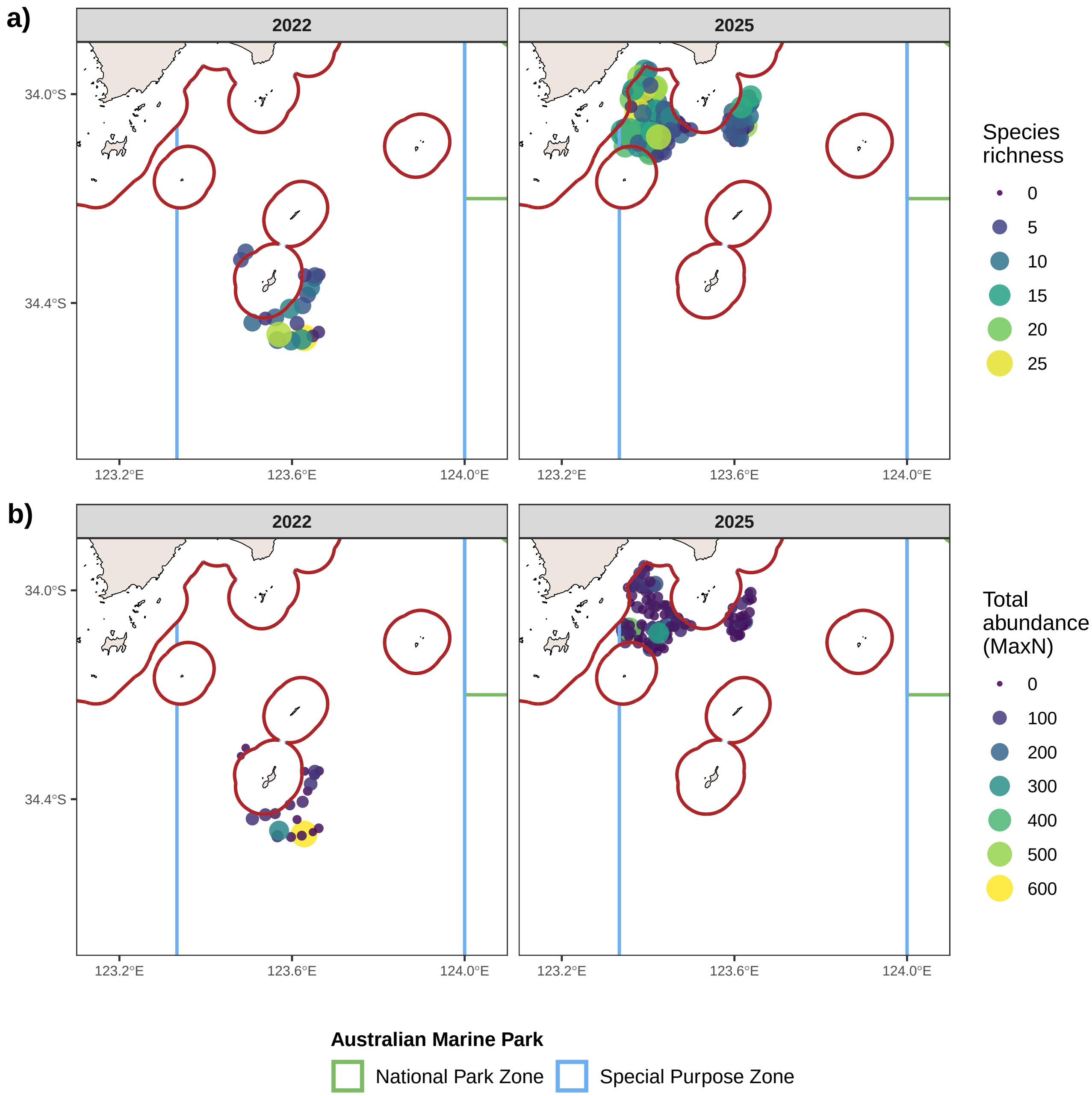

Figure 8: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV deployment. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

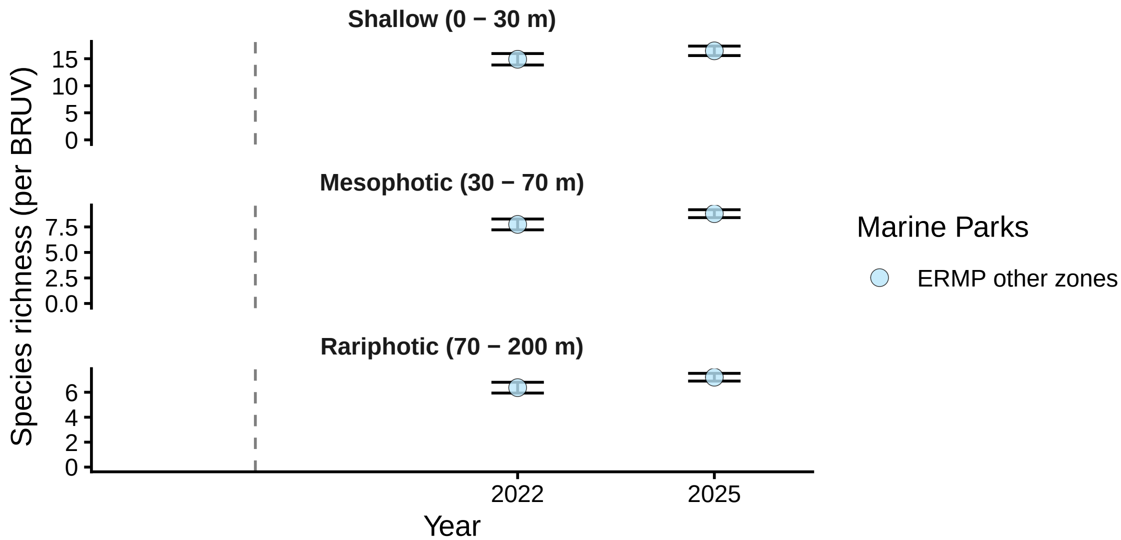

Figure 9.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth).

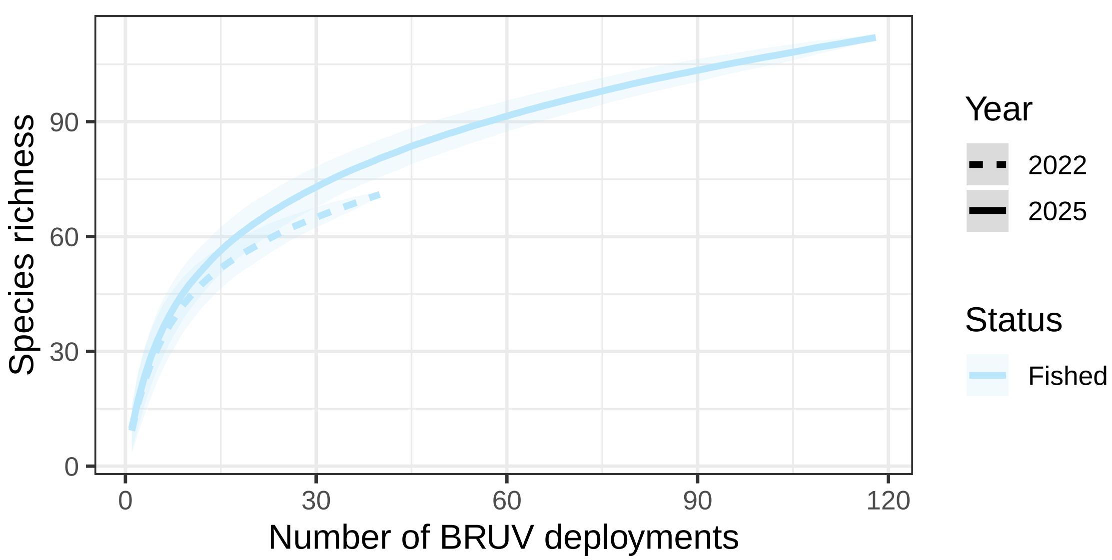

Figure 9.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth).

Species richness

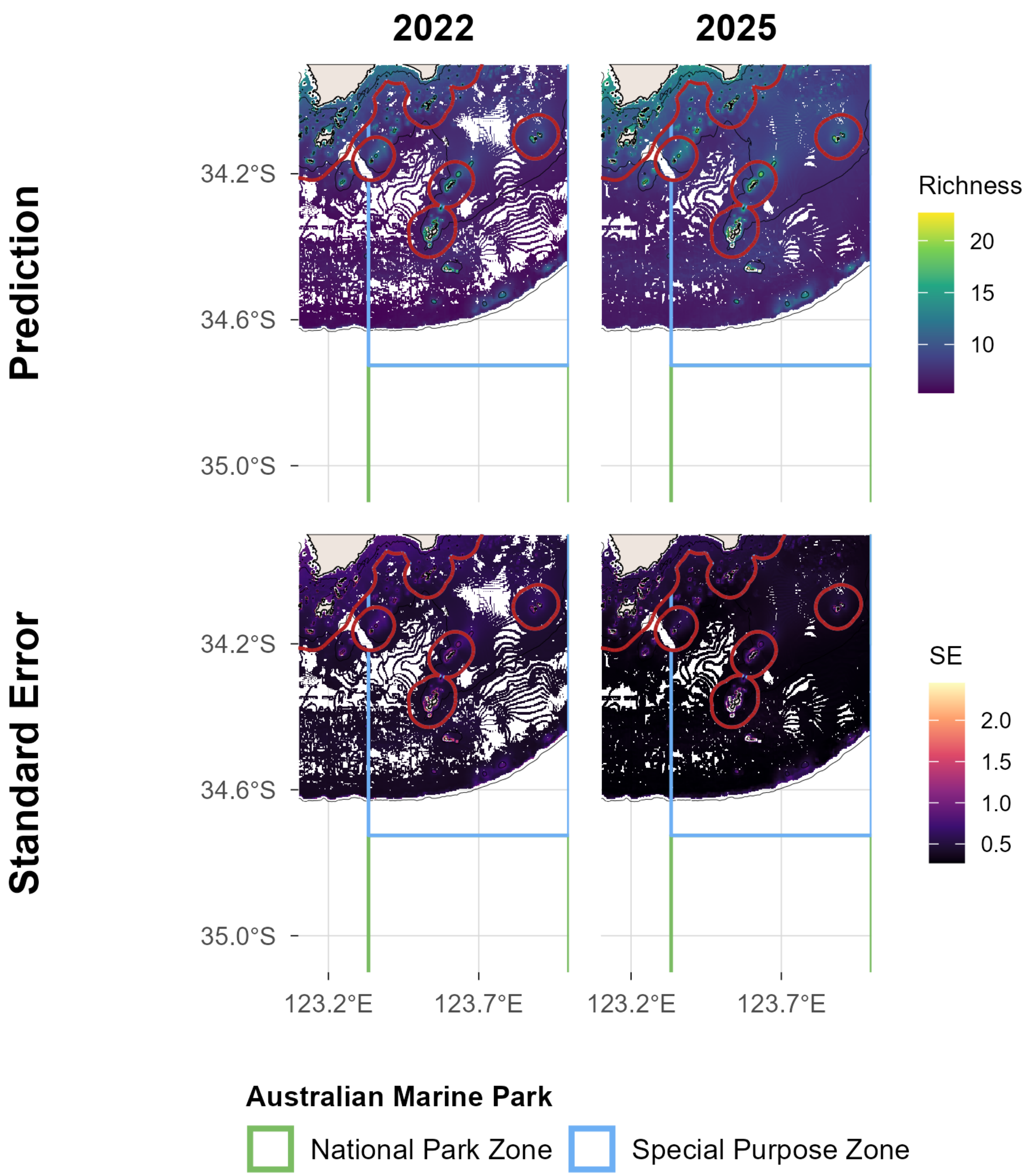

Figure 9.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

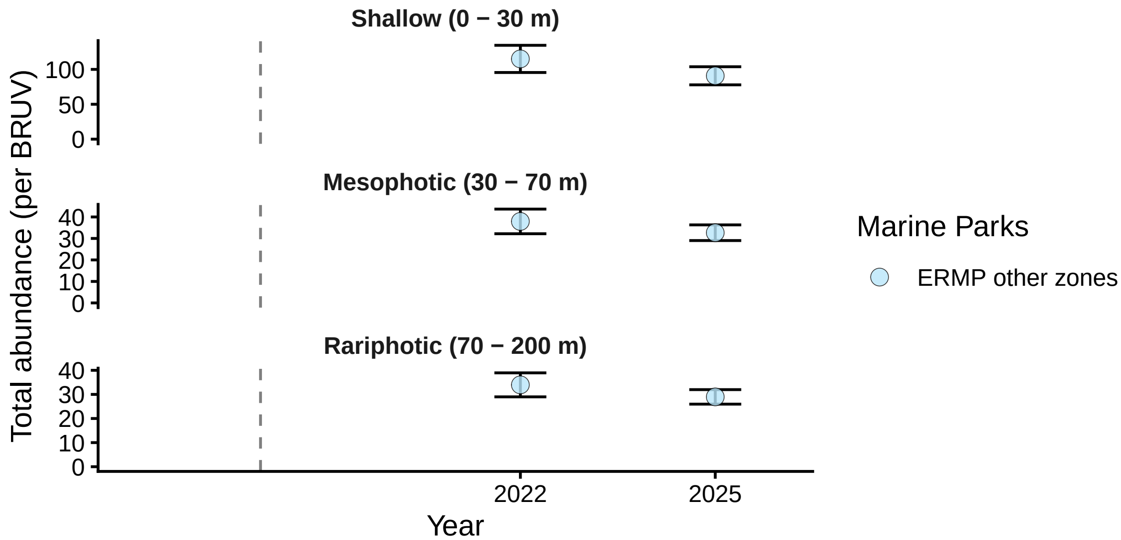

Figure 10.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth).

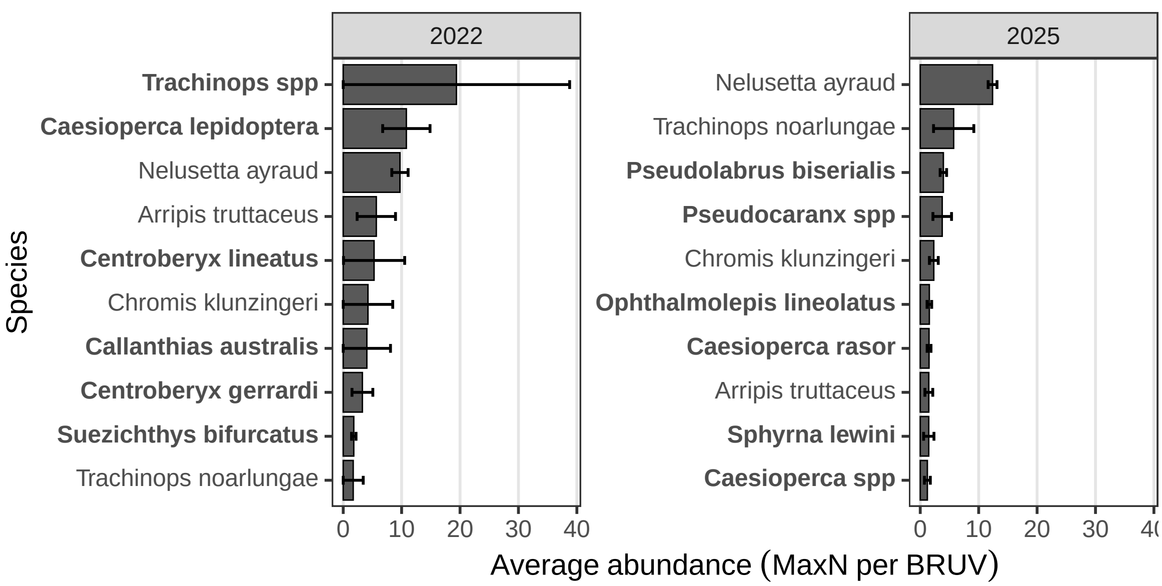

Figure 10.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth).

Total abundance

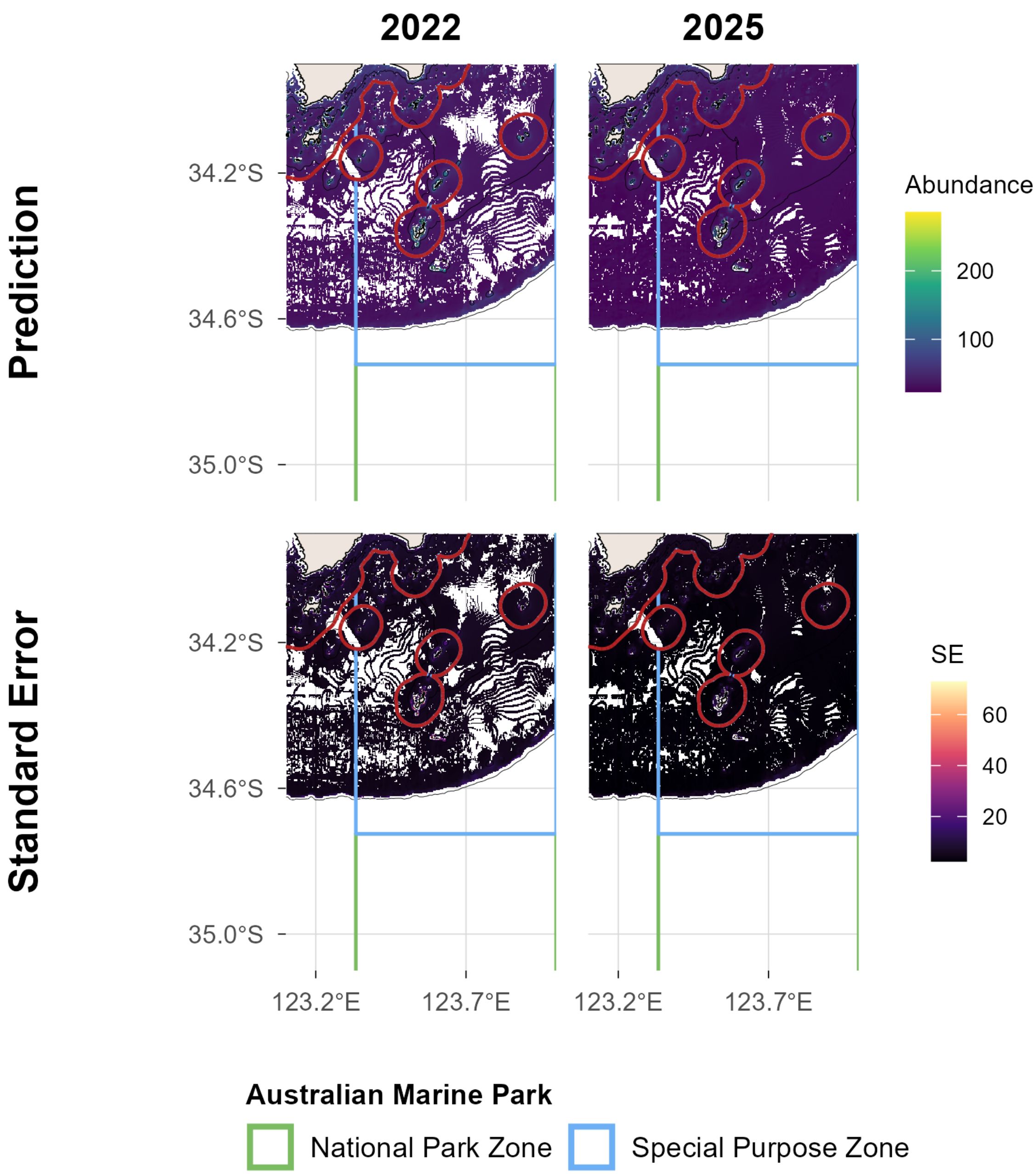

Figure 10.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

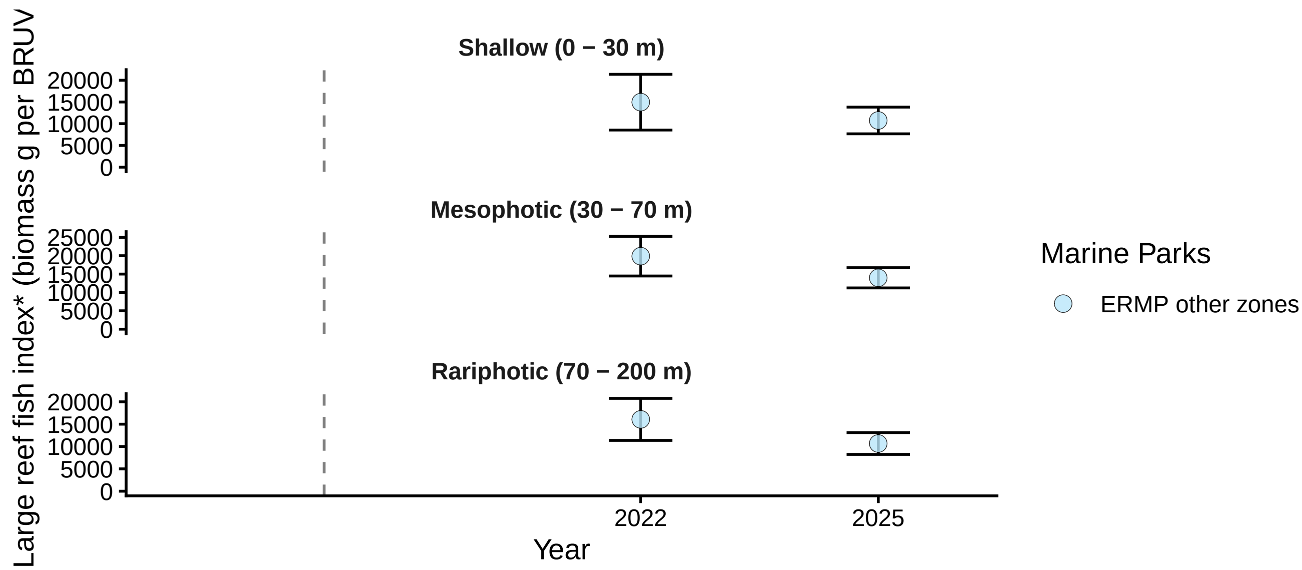

Figure 11.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

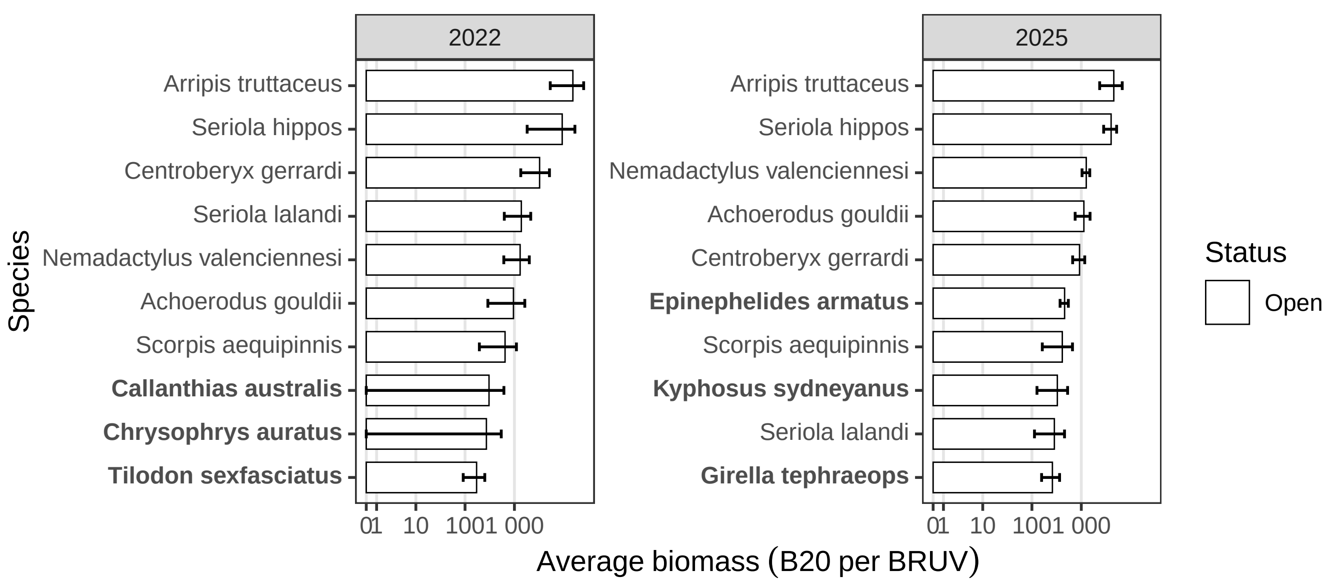

Figure 11.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth).

Large Reef Fish Index*

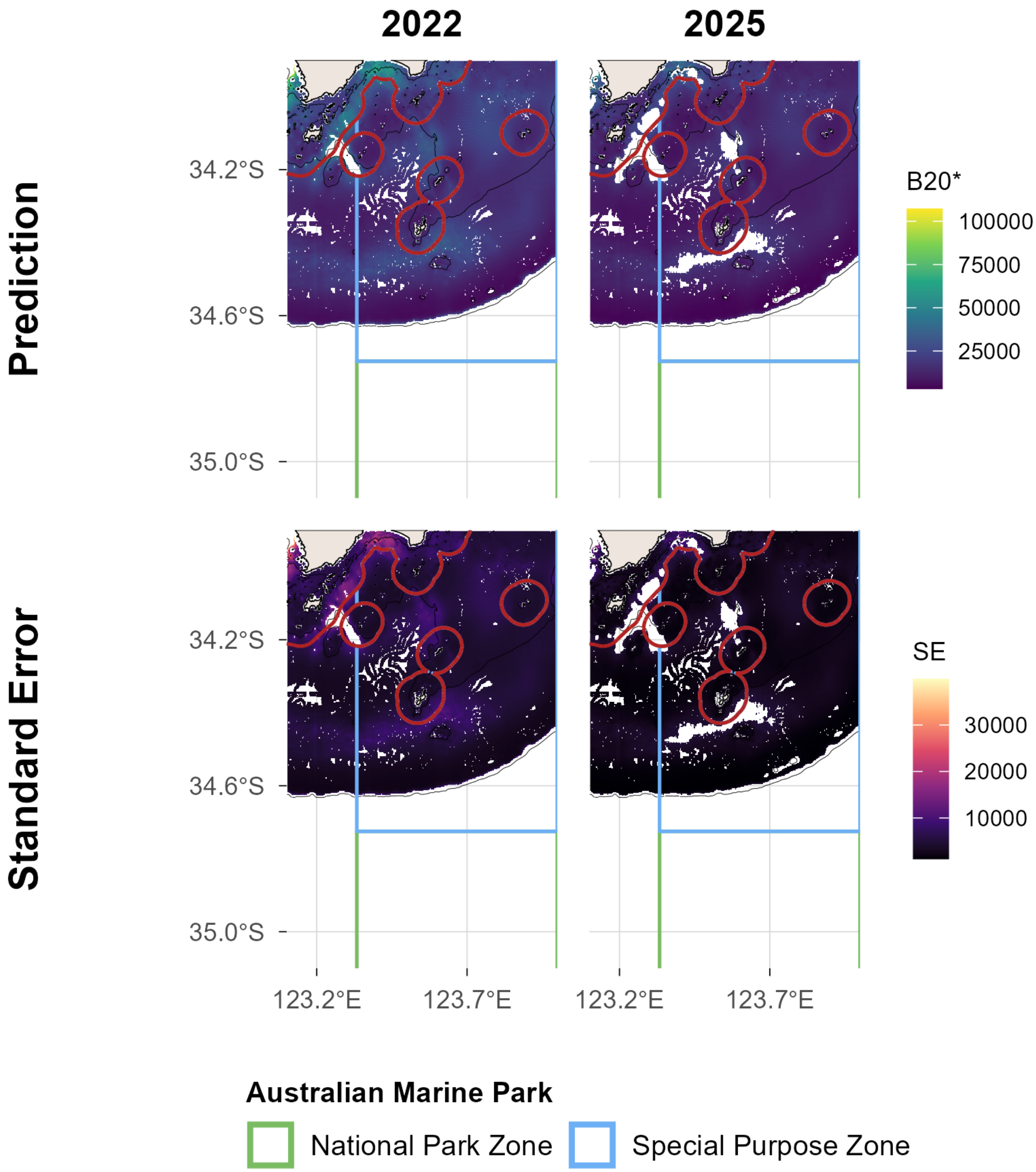

Figure 11.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

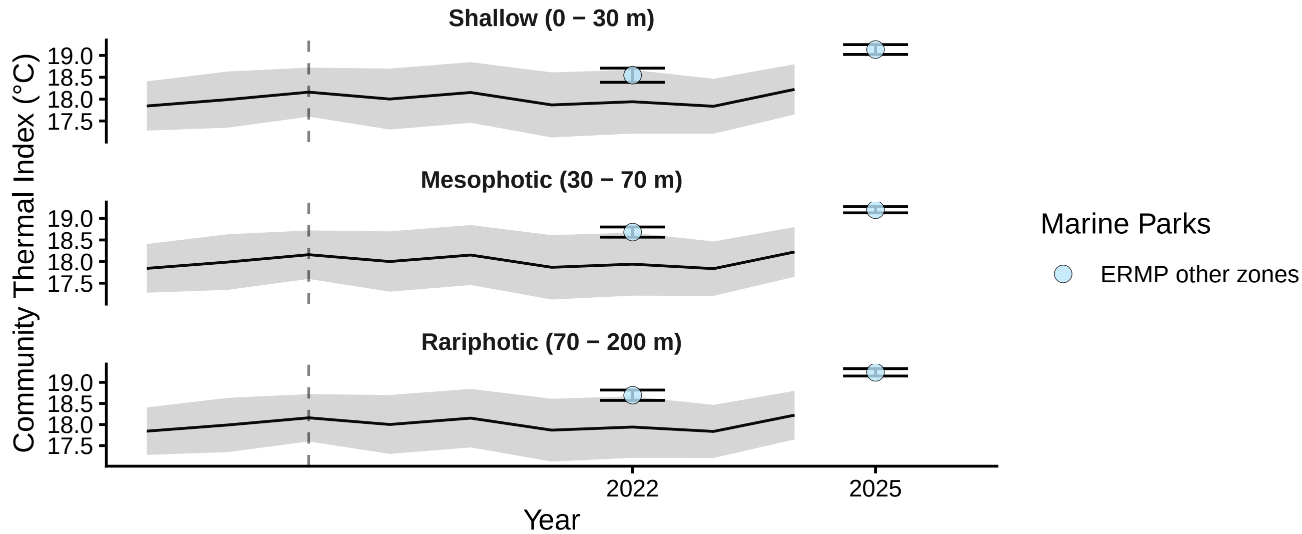

Figure 12.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth).

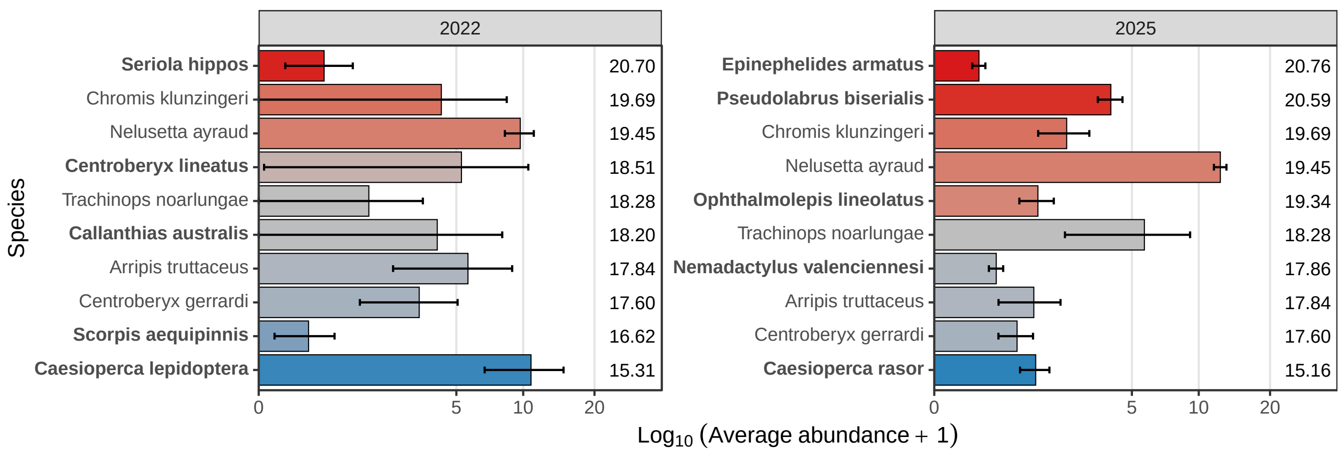

Figure 12.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. ERMP = Eastern Recherche Marine Park (Commonwealth).

Reef fish thermal index

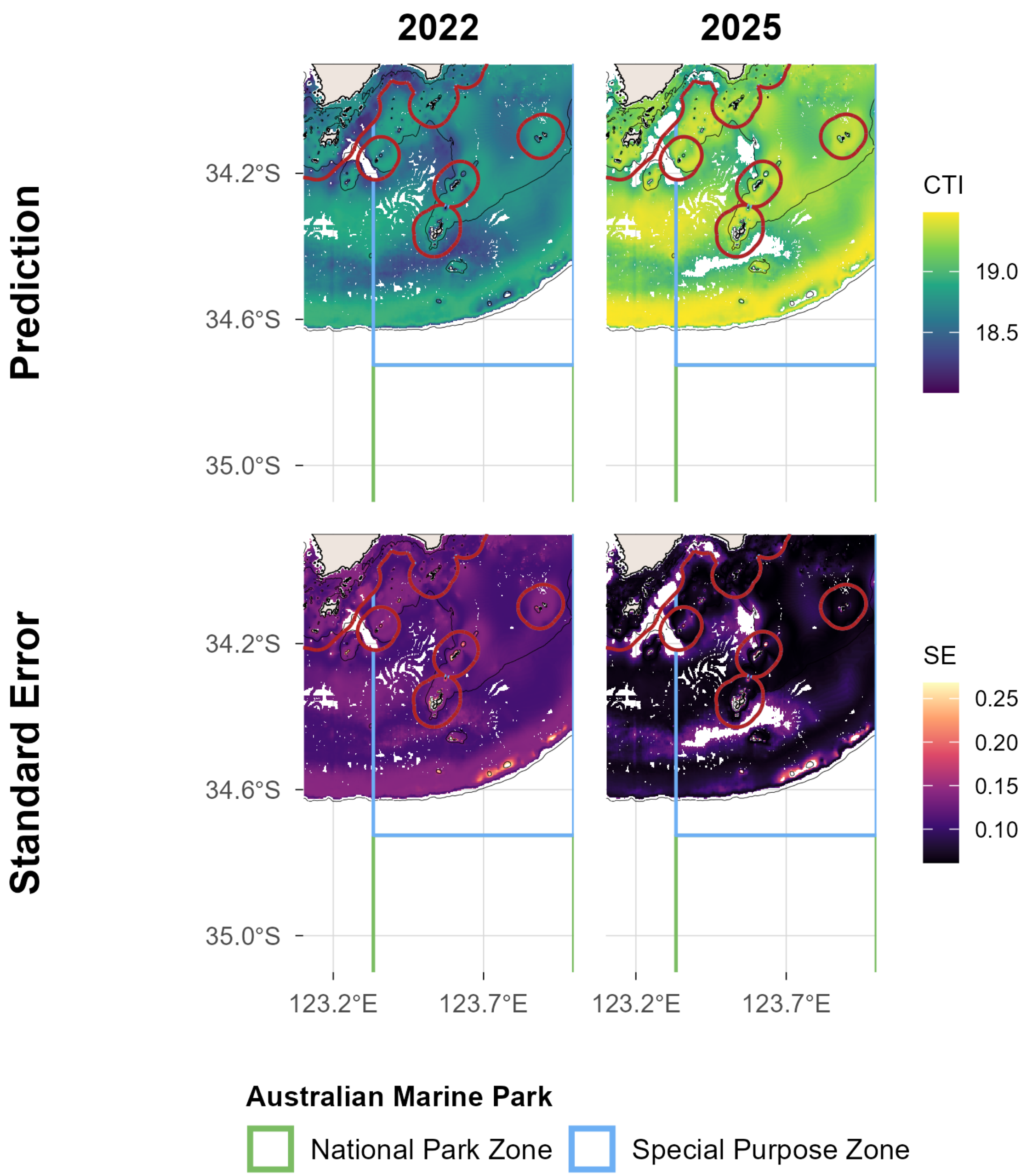

Figure 12.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.